

NARI NIKETAN

COMPLETE TECHNOLOGY, SYSTEM ARCHITECTURE & BACKEND API DOCUMENTATION

Comprehensive Technical Reference Manual & Engineering Audit

Version: 2.2.0 (Production Node.js Backend API Release) | **Date:** September 2026

Platform: High-Performance Indian Ethnic E-Commerce Platform + Cloud Run Microservice

Storefront URL: <https://www.nariniketan.shop> | **Firebase App:** <https://nari-niketan.web.app>

Backend API URL: <https://nari-niketan-api-997712460310.asia-south1.run.app>

Audit Metric / Attribute	Verified Value / State
Architecture Type	Hybrid Jamstack + Node.js/Express Secure API on Google Cloud Run
Frontend Storefront	Vanilla ES6+ JavaScript, Semantic HTML5, Modular CSS3 Architecture
Backend API Layer	Express 4.19, Helmet, Strict CORS, Tiered Rate Limiting, Winston Audit Logs
Database & Cloud Storage	Cloud Firestore (17 Collections) + Firebase Cloud Storage (5MB MIME Rules)
Authentication & RBAC	Firebase Auth ID Tokens + Firebase Custom Claims (Admin, Seller, Owner)
Pricing & Order Engine	Server-Authoritative Calculation: Zero Trust on Client Prices/Subtotals
Internal AI Suite	Nari AI Stylist RAG, NLP Search, Virtual Try-On, Nari AI Ads, Seller Photo Wizard
Continuous Monitoring	Nari AI Website Operations Center (12 Pillars, 8 Viewports Sandbox Testing)
Fulfilment Model	Dual Mode: Direct Home Delivery + Physical Store Pickup (Google Maps Geocoding)
Hosting & Edge CDN	Firebase Hosting (Global Fastly CDN, TLS 1.3) + Cloud Run (asia-south1)

Document Classification: CONFIDENTIAL / OFFICIAL ENGINEERING SPECIFICATION

Part 1 & 2 — Executive Summary & Complete Technology Stack

1.1 System Purpose & Domain:

Nari Niketan is an enterprise-grade ethnic fashion e-commerce ecosystem serving premium Indian ethnic wear (Sarees, Lehengas, Kurtis, Suits, Dupattas, Gowns, Jewellery). The platform implements a secure 3-tier architecture with zero-trust business logic running on Google Cloud Run.

Layer / Domain	Technologies & Libraries	Version & Notes
Frontend UI / Storefront	HTML5 Semantic, CSS3 Custom Properties, Vanilla ES6+ JS	Zero framework dependencies, ultra-fast load
Backend API Runtime	Node.js, Express.js, Docker, Google Cloud Run	Node v20 LTS, asia-south1 region
Backend Security	Helmet, CORS Allowlist, express-rate-limit, Zod	Strict input validation & tiered DDoS mitigation
SDK & Persistence	firebase-admin SDK, Cloud Firestore NoSQL	v12.3.0, Application Default Credentials (ADC)
Client-Side Auth	Firebase Auth (Email, Google OAuth, Phone OTP)	Bearer ID Token attached to API calls
Image & Asset Storage	Firebase Cloud Storage, Fastly Edge CDN	5MB max upload, image/* MIME validation
Geocoding & Maps	OpenStreetMap, Leaflet.js, Google Maps Platform	GPS coordinate capture & store pickup routing
Logging & Monitoring	Winston Structured Logger, Nari Ops Engine v2.0	Audit logging for all orders, roles, payouts

Part 4 & 5 — 3-Tier System Architecture & Cloud Run API Layer

5.1 Architectural Decoupling & Security Boundary:

To guarantee tamper-proof order totals, discount calculations, and role authorizations, all state-mutating operations are decoupled from the browser and handled by the Node.js Cloud Run service:

- **1. Client Presentation Tier:** HTML5/CSS3/JS storefront hosted on Firebase Global CDN. Executes client-side UX estimates, AI Stylist advisor, and sends Bearer ID tokens on checkout.
- **2. Application Security Tier (Cloud Run):** Express API verifying Firebase ID tokens, validating Zod schemas, enforcing tiered rate limits, calculating pricing authoritatively, and setting Custom Claims.
- **3. Cloud Persistence Tier (GCP):** Cloud Firestore managing 17 distinct collections, Cloud Storage housing product images, and Firebase Auth managing user identities.

Part 12 — Backend API Endpoint Reference Matrix

All API endpoints are mounted under /api/* and require Bearer Token authorization unless marked Public:

Method & Endpoint	Auth Level	Description & Security Controls
GET /api/health	Public	Returns service health, version, and environment status
GET /	Public	Root welcome status and documentation pointer
POST /api/orders	Authenticated	Creates order with server-calculated price, stock check & coupon
GET /api/orders/my	Authenticated	Fetches authenticated user's personal order history
GET /api/orders/:id	Authenticated	Gets single order. Customers see self, Admin all, Seller line items
POST /api/coupons/validate	Authenticated	Validates coupon dates, usage limits, minimum order & discount
POST /api/payments/submit-utr	Authenticated	Submits manual UPI UTR. Rejects duplicate UTRs. Status: Pending
GET /api/payments/status/:id	Authenticated	Returns verification status of order payment
GET /api/admin/orders	Admin / Owner	Lists all store orders with status and date filters
PATCH /api/admin/orders/:id/status	Admin / Owner	Updates order lifecycle state and delivery milestones
PATCH /api/admin/orders/:id/verify-payment	Admin / Owner	Manually confirms customer UPI UTR payment
PATCH /api/admin/users/:uid/role	Owner Only	Sets Firebase Custom Claims (admin, seller, owner)
PATCH /api/admin/users/:uid/block	Admin / Owner	Disables/enables user account in Firebase Auth & Firestore
POST /api/admin/coupons	Admin / Owner	Creates new promotional discount coupon
GET /api/seller/products	Seller Role	Lists catalog products belonging to authenticated seller
POST /api/seller/products	Seller Role	Adds new product with sellerId bound from verified token
PATCH /api/seller/products/:id	Seller Role	Updates product details (ownership verified on every update)
GET /api/seller/earnings	Seller Role	Computes revenue breakdown (day, week, month, total)

Part 10 & 11 — Authentication & Firebase Custom Claims RBAC Matrix

11.1 Custom Claims Authorization Model:

Role elevation is strictly enforced via Firebase Auth Custom Claims set by the backend Admin SDK. Clients cannot alter their own claims:

Role Level	Claim Property	Target User Scope	Privilege Scope
Customer / Guest	Default (No Claims)	All registered shoppers	Browse catalog, create orders, view personal orders
Approved Seller	{ seller: true }	Verified vendor partners	Manage own products, view order line items & earnings
Store Admin	{ admin: true }	Operations & support managers	Order management, payment verification, coupon creation
Platform Owner	{ owner: true }	Verified root administrators	Full system access + role elevation & payout dispatch

Part 13 & 14 — Internal AI Suite & Operations Center Sentinel

13.1 Rule-Based Client-Side AI Engines:

All internal AI features operate locally in the browser with zero external paid API dependencies, ensuring privacy, instant latency, and zero token costs:

- **1. Nari AI Stylist Advisor (js/nari-ai-stylist.js):** Multi-criteria outfit recommendation engine utilizing body shape analysis, occasion taxonomy, and color harmony charts.
- **2. Natural Language Search (js/nari-ai-search.js):** Query tokenizer extracting colors, fabrics, prices, and categories into interactive search filter chips.
- **3. Virtual Try-On Canvas (js/nari-virtual-tryon.js):** HTML5 Canvas composite tool overlaying drape silhouettes on customer-uploaded photos.
- **4. Nari AI Ad Studio (js/nari-ai-ads.js):** Contextual ad headline and CTA generator tracking variant click-through performance.
- **5. Seller Copywriter & Photo Assistant (seller/seller-ai.js):** Structured product description generator across 7 brand voices and 6-angle photo checklist.

14.1 Website Operations Center v2.0 (12 Pillars Live Health Monitoring):

Pillar	Inspection Scope	Verification Method
1. Website Routes	Crawls 16 core pages, detects 404s and broken links	Live fetch & sandbox DOM link parsing
2. QA & Journeys	Form constraints, button handlers, Add-to-Cart state	Simulated click events & form checks
3. Multi-Viewport	Overflow checks across 8 viewports (320px–1280px)	Isolated iframe sandbox scrollWidth probe
4. Backend API	Endpoint response codes, latency & Cloud Run status	Health check probe on /api/health
5. Firestore DB	Firestore query latency and collection read/write rules	Live read probe on 'products' collection
6. Security & Config	HTTPS, X-Frame-Options, MIME sniffing, safe scripts	Response header scanner
7. Performance	DOM load time, page weight, heavy images (>500KB)	Performance API & image dimensions
8. SEO & Metadata	Title tags, meta descriptions, single H1, canonical tags	HTML head parser
9. Accessibility	Image alt tags, accessible button names, input labels	WCAG 2.1 AA attribute auditor
10. Seller Portal	6-image upload constraints, draft/publish state	Seller form & image array validator
11. Admin Operations	Order state transitions, invoice calculations	Admin state machine probe
12. AI Services	Availability of Stylist, Search, Ads & Try-on	Global AI object integrity probe

Part 41 — Master System Architecture Blueprint

TIER 1: CLIENT ACCESS TIERS (Presentation Layer)		
Customer Storefront shop.html, cart.html, checkout.html Firebase Hosting (Global Fastly CDN)	Seller Vendor Portal /seller/index.html Catalog, Inventory, AI Copywriter	Admin Operations Portal /admin/index.html Orders, Users, Payouts, Health Sentinel
HTTPS REST Requests + Bearer Firebase ID Token (Authorization Header) ↓		
TIER 2: NODE.JS / EXPRESS BACKEND API (Google Cloud Run — asia-south1)		
Security & Auth Gateway • authenticateFirebaseUser (ID Tokens) • Firebase Custom Claims (admin/seller/owner) • Helmet Security Headers & Strict CORS • Tiered Rate Limiters (DDoS Protection)	Server Business Logic Engine • Zero-Trust Pricing Calculation • Server-Authoritative Coupon Engine • Inventory Stock & Availability Checks • Manual UPI UTR Duplicate Safeguard	Observability & Admin SDK • Winston Structured Audit Logging • Privileged Firebase Admin SDK (ADC) • Zod Request Schema Validation • Global Safe Error Handler
Privileged Firestore & Storage Protocols (Application Default Credentials) ↓		
TIER 3: CLOUD PERSISTENCE & MANAGED SERVICES (Google Cloud Platform)		
Cloud Firestore NoSQL 17 Collections: products, orders, users, coupons, categories, banners, systemAlerts, payouts	Firebase Cloud Storage Product media, seller uploads, 5MB image/* rules, delivery proofs	Firebase Authentication Google OAuth, Email/Password, Phone OTP Identity Store

Part 42 — Final Implementation Verification Checklist

Component / Area	Implementation State	Readiness	Production Verification Notes
Storefront UI & Navigation	Live on Firebase	100% Ready	Zero-overflow across 320px–1280px viewports
Backend API on Cloud Run	Live in asia-south1	100% Ready	Revision 00002-rww serving 100% traffic, /api/health OK
Server Pricing & Orders	Server-Authoritative	100% Ready	Zero trust on client totals; prices fetched from DB
Custom Claims RBAC	Implemented	100% Ready	Admin & Seller roles verified via verified token claims
Manual UPI Verification	Pending Review State	100% Ready	UTR submissions queued for admin approval; dupes blocked
Firestore Security Rules	Deployed & Hardened	100% Ready	Evaluates Custom Claims with zero-leak user rules